SINGLE PHASE A.C. CIRCUITS

- Average value, R.M.S. value, form factor and peak factor for sinusoidal wave form.
- Steady State Analysis of series R-L-C circuits.
- Concept of Reactance, Impedance, Susceptance, Admittance.
- Concept of Power Factor, Real, Reactive and Complex power.
- Illustrative Problems.

RMS VALUE:

- The **RMS** (**Root Mean Square**) value (also known as **effective** or **virtual** value) of of an alternating current (AC) is the value of direct current (DC) when flowing through a circuit or resistor for the specific time period and produces same amount of heat which produced by the alternating current (AC) when flowing through the same circuit or resistor for a specific time.
- The value of an AC which will produce the same amount of heat while passing through in a heating element (such as resistor) as DC produces through the element is called R.M.S Value.
- In short,
- The RMS Value of an Alternating Current is that when it compares to the Direct Current, then both AC and DC current produce the same amount of heat when flowing through the same circuit for a specific time period.

$$I_{RMS} = \frac{I_M}{\sqrt{2}} \;,\; V_{RMS} = \frac{V_M}{\sqrt{2}}$$

For a sinusoidal wave

$$I = 0.707 \times I_M \;,\; V = 0.707 \times V_M$$

,

or

$$\mathbf{I_{RMS} = 0.707 \times I_M \;,\; E_{RMS} = 0.707\, E_M}$$

- Actually, the RMS value of a sine wave is the measurement of heating effect of sine wave. For example, when a resistor is connected to across an AC voltage source, it produces specific amount of heat (Fig 2 – a). When the same resistor is connected across the DC voltage source as shown in (fig 2 – b). By adjusting the value of DC voltage to get the same amount of heat generated before in AC voltage source in fig a. It means the RMS value of a sine wave is equal to the DC Voltage source producing the same amount of heat generated by AC Voltage source.
  - In more clear words, the domestic voltage level in US is 110V, while 220V AC in UK. This voltage level shows the effective value of (110V or 220V R.M.S) and it shows that the home wall socket is capable to provide the same amount of average positive power as 110V or 220V DC Voltage.
  - **Keep in mind that the ampere meters and volt meters connected in AC circuits always showing the RMS values (of current and voltage).**
  - For AC sine wave, RMS values of current and voltage are:

$$\mathbf{I_{RMS} = 0.707 \times I_M \;,\; V_{RMS} = 0.707\, V_M}$$

- Let’s see how to find the R.M.S values of a sine wave.
- We know that the value of sinusoidal alternating current (AC) =

$$\mathbf{I_m \sin \omega\,\theta = I_m \sin \theta}$$

- While the mean of square of instantaneous values of current in in half or complete cycle is:

$$= \int_0^{2\pi} \frac{i^2\, d\theta}{(2\pi - 0)}$$

The Square root of this value is:

$$= \sqrt{\left(\int_0^{2\pi} \frac{i^2\, d\theta}{2\pi} \right)}$$

Hence, the RMS value of the current is (while putting I = I_m Sin θ):

$$I = \sqrt{\left(\int_0^{2\pi} \frac{i^2 d\theta}{2\pi} \right)} = \sqrt{\left(\frac{I_m^2}{2\pi} \int_0^{2\pi} \sin^2\theta\, d\theta \right)}$$

Now,

$$\cos 2\theta = 1 - 2\sin^2\theta \quad \therefore \quad \sin^2\theta = \frac{1 - \cos 2\theta}{2}$$

$$I = \sqrt{\left(\frac{I_m^2}{4\pi} \int_0^{2\pi} (1 - \cos 2\theta)\, d\theta \right)} = \sqrt{\left(\frac{I_m^2}{4\pi} \left| \theta - \frac{\sin 2\theta}{2} \right|_0^{2\pi} \right)}$$

$$= \sqrt{\frac{I_m^2}{4} \quad 2} \qquad \sqrt{\frac{I_m^2}{2}} \qquad \therefore \boxed{I = \frac{I_m}{\sqrt{2}} = 707\ I_m}$$

Therefore, We may find that for a symmetrical sinusoidal current:

I_{RMS} = Max Value of Current x 0.707

Average Value:

If we convert the alternating current (AC) sine wave into direct current (DC) sine wave through rectifiers, then the converted value to the DC is known as the average value of that alternating current sine wave.

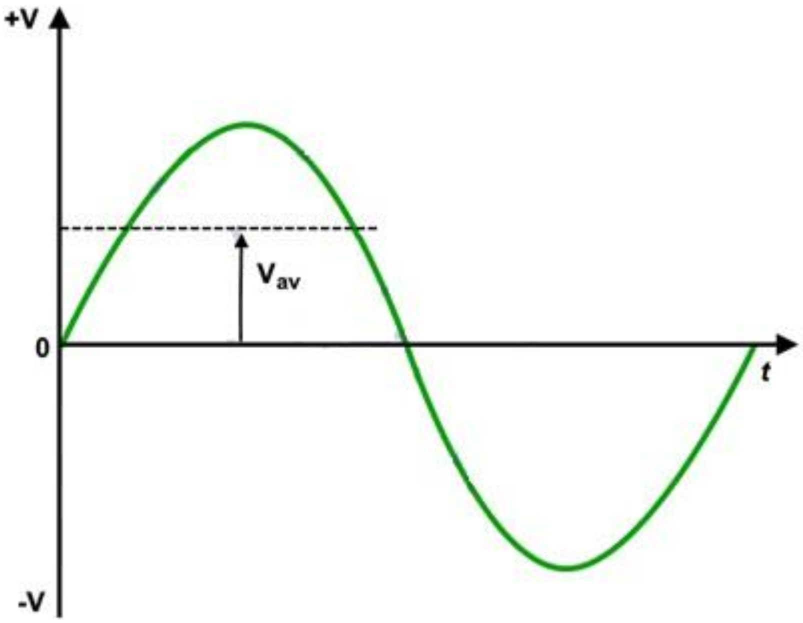


Fig 4 – Average Value of Voltage

If the maximum value of alternating current is "I_{MAX}", then the value of converted DC current through rectifier would be "0.637 I_M" which is known as average value of the AC Sine wave (I_{AV}).

Average Value of Current = I_{AV} = 0.637 I_M

Average Value of Voltage = E_{AV} = 0.637 E_M

The Average Value (also known as Mean Value) of an Alternating Current (AC) is expressed by that Direct Current (DC) which transfers across any circuit the same amount of charge as is transferred by that Alternating Current (AC) during the same time.

Keep in mind that the average or mean value of a full sinusoidal wave is "Zero" the value of current in first half (Positive) is equal to the the next half cycle (Negative) in the opposite direction. In other words, There are same amount of current in the positive and negative half cycles which flows in the opposite direction, so the average value for a complete sine wave would be "0". That's the reason that's why we don't use average value for plating and battery charging. If an AC wave is converted into DC through a rectifier, It can be used for electrochemical works.

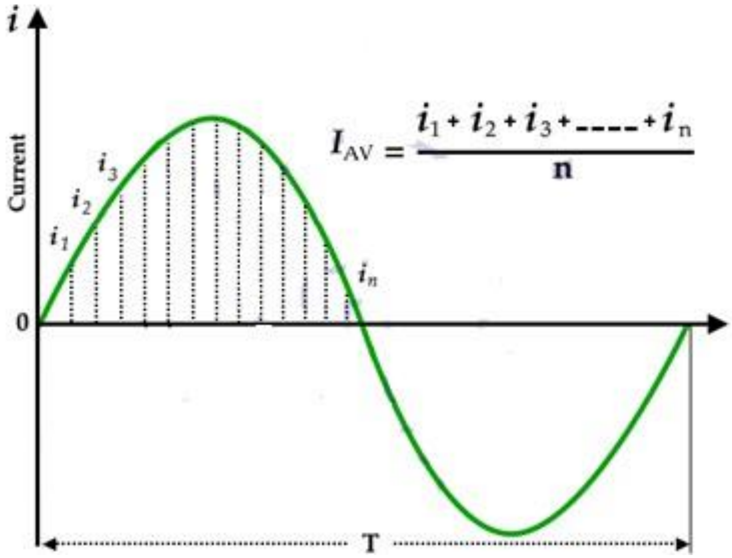


Fig 5- Average Value of Current

In short, the average value of a sine wave taken over a complete cycle is always zero, because the positive values (above the zero crossing) offset or neutralize the negative values (below the zero crossing.)

We know that the standard equation of alternating current is

$$\mathbf{i = Sin\ \omega\ \theta = I_m\, Sin\ \theta}$$

- Maximum value of current on sine wave = I_m
- Average value of current on sine wave = I_{AV}
- Instantaneous value of current on sine wave = *i*
- The angle specified fir "*i*" after zero position of current = θ
- Angle of half cycle = π radians
- Angle of full cycle = 2π radians

(a) Average value of complete cycle:

Let i = Sin ω θ = I_m Sin θ

$$I_{AV} = \frac{1}{2\pi}\int_0^{2\pi} i\,d\theta = \frac{1}{2\pi}\int_0^{2\pi} i_m \,\text{Sin}\theta\, d\theta$$

$$= \frac{I_m}{2\pi}\left[-\text{Cos}\,\theta\right]_0^{2\pi} = \frac{I_m}{2\pi}(\text{Cos}\,2\pi - \text{Cos}\,0)$$

$$= \frac{I_m}{2\pi}(1-1) = 0\,; \quad I_{AV} = 0$$

Thus, the average value of a sinusoidal wave over a complete cycle is zero.

(b) Average value of current over a half cycle

$$I_{AV} = \frac{1}{\pi}\int_0^{\pi} i\, d\theta$$

$$= \frac{1}{\pi}\int_0^{\pi} I_m \,\text{Sin}\,\theta . d\theta \quad [\because i = I_m\,\text{Sin}\,\theta]$$

$$= \frac{I_m}{\pi}\left[-\text{Cos}\,\theta\right]_0^{\pi} = \frac{I_m}{\pi}\left[-\text{Cos}\,\pi - (-\text{Cos}\,0)\right]$$

$$= \frac{I_m}{\pi}[(+1) - (-1)] = \frac{I_m}{\pi}(+2)$$

$$I_{AV} = \frac{2}{\pi} I_m = 0.637\ A$$

Average Value of Current (Half Cycle)

$$\mathbf{I_{AV} = 0.637\ V_M}$$

Similarly, the average value of voltage over a half cycle

$$\mathbf{V_{AV} = 0.637\ V_M}$$

★ **Average Voltage Value**

$$V_{AV} = \frac{2V_P}{\pi} = 0.637 \times V_P$$

$$V_{AV} = \frac{2V_M}{\pi} = 0.637 \times V_M$$

★ **Average Current Value**

$$I_{AV} = \frac{2I_M}{\pi} = 0.637 \times I_M$$

What is Peak Voltage or Maximum Voltage Value ?

Peak value is also known as **Maximum Value**, **Crest Value** or **Amplitude**. It is the maximum value of alternating current or voltage from the “0” position no matter positive or negative half cycle in a sinusoidal wave as shown in fig 8. Its expressed as $\mathbf{I_M}$ and $\mathbf{E_M}$ or $\mathbf{V_P}$ and $\mathbf{I_M}$.

Equations of Peak Voltage Value is:

$$\mathbf{V_P = \sqrt{2} \times V_{RMS} = 1.414\ V_{RMS}}$$

$$\mathbf{V_P = V_{P\text{-}P}/2 = 0.5\ V_{P\text{-}P}}$$

$$\mathbf{V_P = \pi/2 \times V_{AV} = 1.571 \times V_{AV}}$$

In other words, It is the value of voltage or current at the positive or the negative maximum (peaks) with respect to zero. In simple words, it is the instantaneous value with maximum intensity.

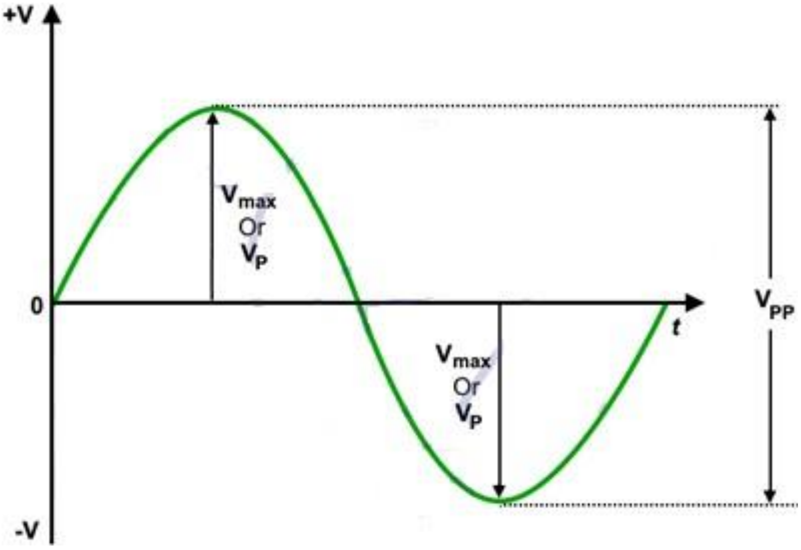


Fig 8 – Peak or Maximum Values of Voltages

Peak to Peak Value:

The sum of positive and negative peak values is known as peak to peak value. Its expressed as $\mathbf{I_{PP}}$ or $\mathbf{V_{PP}}$.

Equations and formulas for Peak to Peak Voltage are as follow:

$$\mathbf{V_{P\text{-}P} = 2\sqrt{2}} \times \mathbf{V_{RMS} = 2.828} \times \mathbf{V_{RMS}}$$

$$\mathbf{V_{P\text{-}P} = 2} \times \mathbf{V_P}$$

$$\mathbf{V_{P\text{-}P} = \pi \times V_{AV} = 3.141} \times \mathbf{V_{AV}}$$

In other words, the peak to peak value of a sine wave, is the voltage or current from positive peak to the negative peak and its value is double as compared to peak value or maximum value as shown in fig 8 above.

Peak Factor:

Peak Factor is also known as Crest Factor or Amplitude Factor.

It is the ratio between maximum value and RMS value of an alternating wave.

$$\textbf{Peak Factor} = \frac{\text{Maximum Value}}{\text{R.M.S Value}}$$

For a sinusoidal alternating voltage:

$$\frac{E_M}{0.707\,E_M} = \mathbf{1.414}$$

For a sinusoidal alternating current:

$$\frac{I_M}{0.707\, I_M} = \mathbf{1.414}$$

Form Factor:

The ratio between RMS value and Average value of an alternating quantity (Current or Voltage) is known as Form Factor.

$$\textbf{Form Factor} = \frac{\text{RMS Value}}{\text{Average Value}}$$

$$= \frac{0.707\, E_M}{0.637\, E_M} \text{ Or } \frac{0.707\, I_M}{0.637\, I_M} = \mathbf{1.11}$$

Other Terms Related To AC Circuits

Waveform

- The path traced by a quantity (such as voltage or current) plotted as a function of some variable (such as time, degree, radians, temperature etc.) is called waveform.

Cycle

1. One complete set of positive and negative values of alternating quality (such as voltage and current) is known as cycle.
2. The portion of a waveform contained in one period of time is called cycle.
3. A distance between two same points related to value and direction is known as cycle.
4. A cycle is a complete alternation.

Period

- The time taken by a alternating quantity (such as current or voltage) to complete one cycle is called its time period "T".
- It is inversely proportional to the Frequency "*f*" and denoted by "T" where the unit of time period is second.
- Mathematically;

$$\mathbf{T = 1/\mathit{f}}$$

Frequency

- Frequency is the number if cycles passed through per second. It is denoted by "f" and has the unit cycle per second i.e. Hz (Herts).
- The number of completed cycles in 1 second is called frequency.
- It is the number of cycles of alternating quantity per second in hertz.
- Frequency is the number of cycles that a sine wave completed in one second or the number of cycles that occurs in one second.

$$\mathbf{\mathit{f} = 1/T}$$

Amplitude

- The maximum value, positive or negative, of an alternating quantity such as voltage or current is known as its amplitude. Its denoted by V_P, I_P or E_{MAX} and I_{MAX}.
- Alternation
- One half cycle of a sine wave (Negative or Positive) is known as alternation which span is 180° degree.

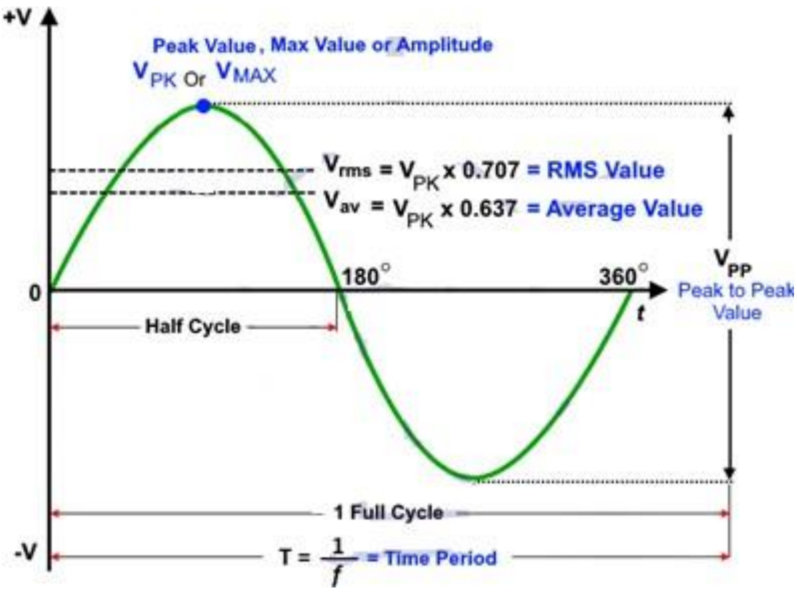


Fig 9 – Different Terms used in AC Circuits and Sine Wave

Introduction to Single Phase AC Circuit:

- In a dc circuit the relationship between the applied voltage V and current flowing through the circuit I is a simple one and is given by the expression I = V/R but in an a c circuit this simple relationship does not hold good. Variations in current and applied voltage set up magnetic and electrostatic effects respectively and these must be taken into account with the resistance of the circuit while determining the quantitative relations between current and applied voltage.
- With comparatively low-voltage, heavy- current circuits magnetic effects may be very large, but electrostatic effects are usually negligible. On the other hand with high-voltage circuits electrostatic effects may be of appreciable magnitude, and magnetic effects are also present.
- Here it has been discussed how the magnetic effects due to variations in current do and electrostatic effects due to variations in the applied voltage affect the relationship between the applied voltage and current.

Purely Resistive Circuit:

- A purely resistive or a non-inductive circuit is a circuit which has inductance so small that at normal frequency its reactance is negligible as compared to its resistance. Ordinary filament lamps, water resistances etc., are the examples of non-inductive resistances. If the circuit is purely non-inductive, no reactance emf (i.e., self- induced or back emf) is set up and whole of the applied voltage is utilized in overcoming the ohmic resistance of the circuit.
- Consider an ac circuit containing a non-inductive resistance of R ohms connected across a sinusoidal voltage represented by v = V sin wt, as shown in Fig.

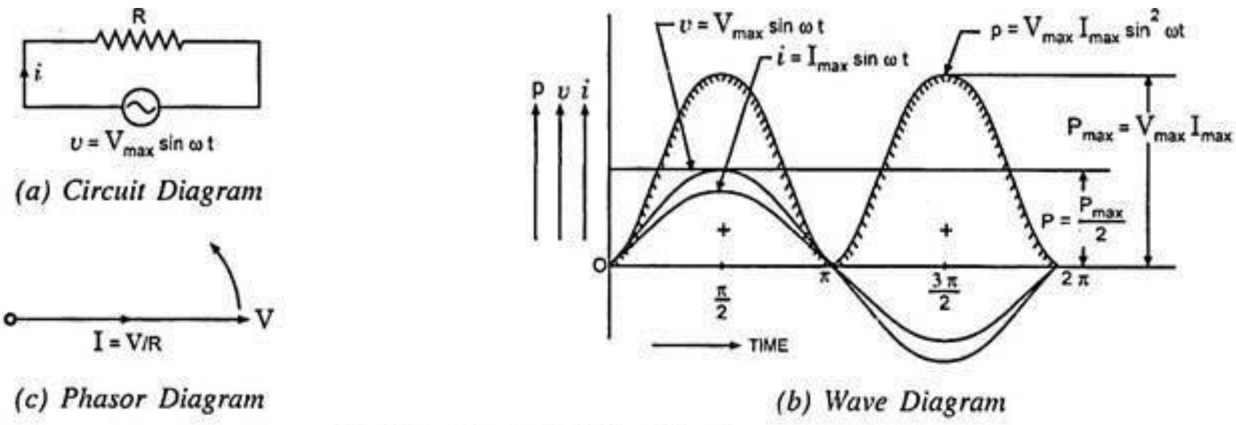


Fig. 4.1 *Purely Resistive Circuit*

As already said, when the current flowing through a pure resistance changes, no back emf is set up, therefore, applied voltage has to overcome the ohmic drop of i R only:

$$i.e. \quad i\,R = v$$

$$\text{or} \quad i = \frac{v}{R} = \frac{V_{max}}{R} \sin \omega t$$

Current will be maximum when $\omega t = \frac{\pi}{2}$ or $\sin \omega t = 1$

$$\therefore \quad I_{max} = \frac{V_{max}}{R}$$

And instantaneous current may be expressed as:
$i = I_{max} \sin \omega t$
From the expressions of instantaneous applied voltage and instantaneous current, it is evident that in a pure resistive circuit, the applied voltage and current are in phase with each other, as shown by wave and phasor diagrams in Figs. 4.1 (b) and (c) respectively.

Power in Purely Resistive Circuit:

The instantaneous power delivered to the circuit in question is the product of the instantaneous values of applied voltage and current.

$$i.e. \quad p = v\,i = V_{max} \sin \omega t \; I_{max} \sin \omega t = V_{max} \; I_{max} \sin^2 \omega t$$

$$\text{or} \quad p = \frac{V_{max} \; I_{max}}{2} (1 - \cos 2\,\omega\, t) \qquad \text{Since } \sin^2 \omega\, t = \frac{1 - \cos 2\,\omega\, t}{2}$$

$$= \frac{V_{max} \; I_{max}}{2} - \frac{V_{max} \; I_{max}}{2} \cos 2\,\omega\, t$$

$$\text{Average power, } P = \text{Average of } \frac{V_{max} \; I_{max}}{2} - \textit{average of } \frac{V_{max} \; I_{max}}{2} \cos 2\,\omega\, t$$

Since average of $\frac{V_{max} \; I_{max}}{2} \cos 2\,\omega\, t$ over a complete cycle is zero,

$$P = \frac{V_{max} \; I_{max}}{2} = \frac{V_{max}}{\sqrt{2}} \cdot \frac{I_{max}}{\sqrt{2}} = V\,I \text{ watts}$$

Where V and I are the rms values of applied voltage and current respectively.
Thus for purely resistive circuits, the expression for power is the same as for dc circuits. From the power curve for a purely resistive circuit shown in Fig. 4.1 (b) it is evident that power consumed in a pure resistive circuit is not constant, it is fluctuating.
However, it is always positive. This is so because the instantaneous values of voltage and current are always either positive or negative and, therefore, the product is always positive. This means that the voltage source constantly delivers power to the circuit and the circuit consumes it.

Purely Inductive Circuit:

An inductive circuit is a coil with or without an iron core having negligible resistance. Practically pure inductance can never be had as the inductive coil has always small resistance. However, a coil of thick copper wire wound on a laminated iron core has negligible resistance arid is known as a choke coil.
When an alternating voltage is applied to a purely inductive coil, an emf, known as self-induced emf, is induced in the coil which opposes the applied voltage. Since coil has no resistance, at every instant applied voltage has to overcome this self-induced emf only.

Let the applied voltage $v = V_{max} \sin \omega t$

and self inductance of coil = L henry

Self induced emf in the coil, $e_L = -L \dfrac{di}{dt}$

Since applied voltage at every instant is equal and opposite to the self induced emf *i.e.* $v = -e_L$

$$\therefore \quad V_{max} \sin \omega t = -\left(-L\frac{di}{dt}\right)$$

$$\text{or} \quad di = \frac{V_{max}}{L} \sin \omega t \, dt$$

Integrating both sides we get

$$i = \frac{V_{max}}{L} \int \sin \omega t \, dt = \frac{V_{max}}{\omega L} (-\cos \omega t) + A$$

where A is a constant of integration, which is found to be zero from initial conditions

$$i.e. \quad i = \frac{-V_{max}}{\omega L} \cos \omega t = \frac{V_{max}}{\omega L} \sin\left(\omega t - \frac{\pi}{2}\right)$$

Current will be maximum when $\sin\left(\omega t - \frac{\pi}{2}\right) = 1$, hence, maximum value of current,

$$I_{max} = \frac{V_{max}}{\omega L}$$

and instantaneous current may be expressed as $i = I_{max} \sin\left(\omega t - \frac{\pi}{2}\right)$

From the expressions of instantaneous applied voltage and instantaneous current flowing through a purely inductive coil it is observed that the current lags behind the applied voltage by $\pi/2$ as shown in Fig. 4.2 (b) by wave diagram and in Fig 4.2 (c) by phasor diagram.

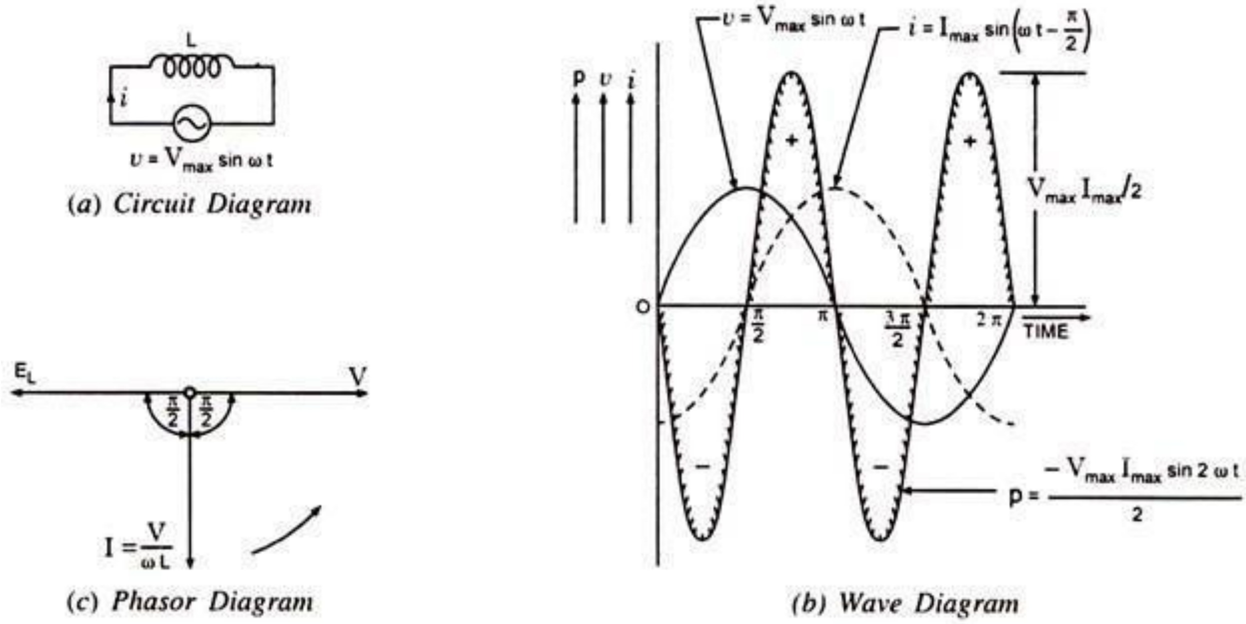


Fig. 4.2 *Purely Inductive Circuit*

Inductive Reactance:

ωL in the expression Imax = $V_{max}/\omega L$ is known as inductive reactance and is denoted by X_L i.e., $X_L = \omega L$
If L is in henry and co is in radians per second then X_L will be in ohms.

Power in Purely Inductive Circuit:

Instantaneous power, $p = v \times i = V_{max} \sin \omega t \, I_{max} \sin (\omega t - \pi/2)$
Or $p = - V_{max} I_{max} \sin \omega t \cos \omega t = V_{max} I_{max}/2 \sin 2 \omega t$
The power measured by wattmeter is the average value of p which is zero since average of a sinusoidal quantity of double frequency over a complete cycle is zero. Hence in a purely inductive circuit power absorbed is zero.

Physically the above fact can be explained as below:

During the second quarter of a cycle the current and the magnetic flux of the coil increases and the coil draws power from the supply source to build up the magnetic field (the power drawn is positive and the energy drawn by the coil from the supply source is represented by the area between the curve p and the time axis). The energy stored in the magnetic field during build up is given as $W_{max} = 1/2 \, L \, I^2_{max}$.
In the next quarter the current decreases. The emf of self-induction will, however, tends to oppose its decrease. The coil acts as a generator of electrical energy, returning the stored energy in the magnetic field to the supply source (now the power drawn by the coil is negative and the curve p lies below the time axis). The chain of events repeats itself during the next half cycles. Thus, a proportion of power is continually exchanged between the field and the inductive circuit and the power consumed by a purely inductive coil is zero.

Purely Capacitive Circuit:

When a dc voltage is impressed across the plates of a perfect condenser, it will become charged to full voltage almost instantaneously. The charging current will flow only during the period of “build up” and will cease to flow as soon as the capacitor has attained the steady voltage of the source. This implies that for a direct current, a capacitor is a break in the circuit or an infinitely high resistance.
In Fig. 4.4 a sinusoidal voltage is applied to a capacitor. During the first quarter-cycle, the applied voltage increases to the peak value, and the capacitor is charged to that value. The current is maximum in the

beginning of the cycle and becomes zero at the maximum value of the applied voltage, so there is a phase difference of 90° between the applied voltage and current. During the first quarter-cycle the current flows in the normal direction through the circuit; hence the current is positive.
In the second quarter-cycle, the voltage applied across the capacitor falls, the capacitor loses its charge, and current flows through it against the applied voltage because the capacitor discharges into the circuit. Thus, the current is negative during the second quarter-cycle and attains a maximum value when the applied voltage is zero.

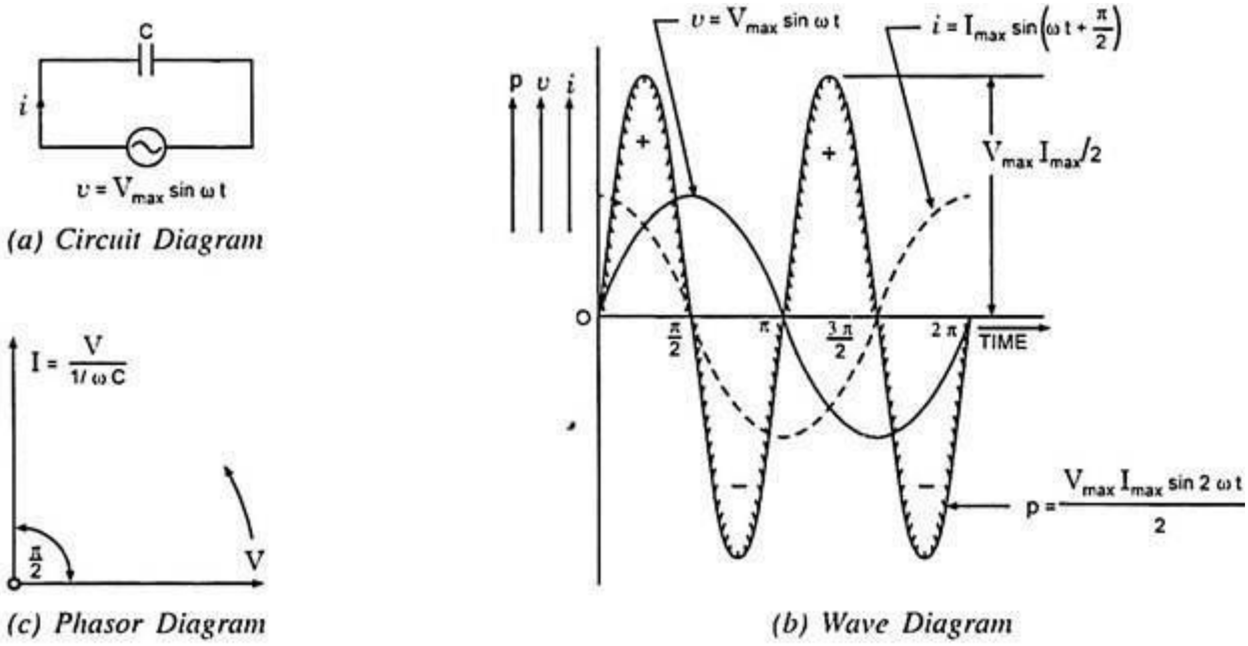


Fig. 4.4 *Purely Capacitive Circuit*

The third and fourth quarter-cycles repeat the events of the first and second, respectively, with the difference that the polarity of the applied voltage is reversed, and there are corresponding current changes.
In other words, an alternating current flow in the circuit because of the charging and discharging of the capacitor. As illustrated in Figs. 4.4 (b) and (c) the current begins its cycle 90 degrees ahead of the voltage, so the current in a capacitor leads the applied voltage by 90 degrees – the opposite of the inductance current-voltage relationship.
Let an alternating voltage represented by $v = V_{max} \sin \omega t$ be applied across a capacitor of capacitance C farads.

The expression for instantaneous charge is given as:

$$q = C V_{max} \sin \omega t$$

Since the capacitor current is equal to the rate of change of charge, the capacitor current may be obtained by differentiating the above equation:

$$i = \frac{dq}{dt} = [C\,V_{max} \sin \omega t] = \omega C\,V_{max} \cos \omega t = \frac{V_{max}}{1/\omega C} \sin\left(\omega t + \frac{\pi}{2}\right)$$

Current is maximum when $t = 0$

$$\therefore \quad I_{max} = \frac{V_{max}}{1/\omega C}$$

Substituting $\frac{V_{max}}{1/\omega C} = I_{max}$ in the above equation for instantaneous current, we get

$$i = I_{max} \sin\left(\omega t + \frac{\pi}{2}\right)$$

From the equations of instantaneous applied voltage and instantaneous current flowing through capacitance, it is observed that the current leads the applied voltage by $\pi/2$, as shown in Figs. 4.4 (b) and (c) by wave and phasor diagrams respectively.

Capacitive Reactance:

$1/\omega C$ in the expression $I_{max} = V_{max}/1/\omega C$ is known as capacitive reactance and is denoted by X_C i.e.,

$$X_C = 1/\omega C$$

If C is in farads and ω is in radians/s, then X_c will be in ohms.

Power in Purely Capacitive Circuit:

$$p = v\,i = V_{max} \sin \omega t.\ I_{max} \sin\left(\omega t + \frac{\pi}{2}\right) = V_{max}\,I_{max} \sin \omega t \cos \omega t$$

$$= \frac{V_{max}\,I_{max}}{2} \sin 2\omega t$$

Average power, $P = \frac{V_{max}\,I_{max}}{2} \times$ average of $\sin 2\,\omega t$ over a complete cycle $= 0$.

Hence power absorbed in a purely capacitive circuit is zero. The same is shown graphically in Fig. 4.4 (b). The energy taken from the supply circuit is stored in the capacitor during the first quarter- cycle and returned during the next.

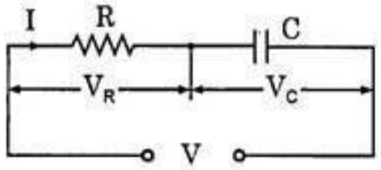
The energy stored by a capacitor at maximum voltage across its plates is given by the expression:

$$W_C = \frac{1}{2}\,C\,V^2_{max}$$

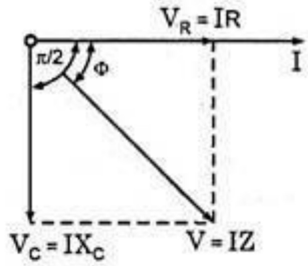
This can be realized when it is recalled that no heat is produced and no work is done while current is flowing through a capacitor. As a matter of fact, in commercial capacitors, there is a slight energy loss in the dielectric in addition to a minute I^2 R loss due to flow of current over the plates having definite ohmic resistance.

The power curve is a sine wave of double the supply frequency. Although it raises the power factor from zero to 0.002 or even a little more, but for ordinary purposes the power factor is taken to be zero. Obviously the phase angle due to dielectric and ohmic losses decreases slightly.

Resistance — Capacitance (R-C) Series Circuit:
Consider an ac circuit consisting of resistance of R ohms and capacitance of C farads connected in series, as shown in Fig. 4.18 (a).
Let the supply frequency be of *f* Hz and current flowing through the circuit be of I amperes (rms value). Voltage drop across resistance, $V_R = I R$ in phase with the current.
Voltage drop across capacitance, $V_C = I X_C$ lagging behind I by π/2 radians or 90°, as shown in Fig. 4.18 (b).



(a) Circuit Diagram



(b) Phasor Diagram

Fig. 4.18

The applied voltage, being equal to phasor sum of V_R and V_C, is given in magnitude by-

$$V = \sqrt{(V_R)^2 + (V_C)^2} = \sqrt{(IR)^2 + (IX_C)^2} = I\sqrt{R^2 + X_C^2} = IZ$$

where $Z^2 = R^2 + X_C^2$

The applied voltage lags behind the current by an angle ϕ:

$$\text{where} \qquad \tan\Phi = \frac{V_C}{V_R} = \frac{IX_C}{IR} = \frac{X_C}{R} = \frac{1}{\omega RC} \text{ or } \Phi = \text{Tan}^{-1}\frac{1}{R\omega C}$$

$$\text{Power factor, } \cos\Phi = \frac{R}{Z}$$

If instantaneous voltage is represented by:

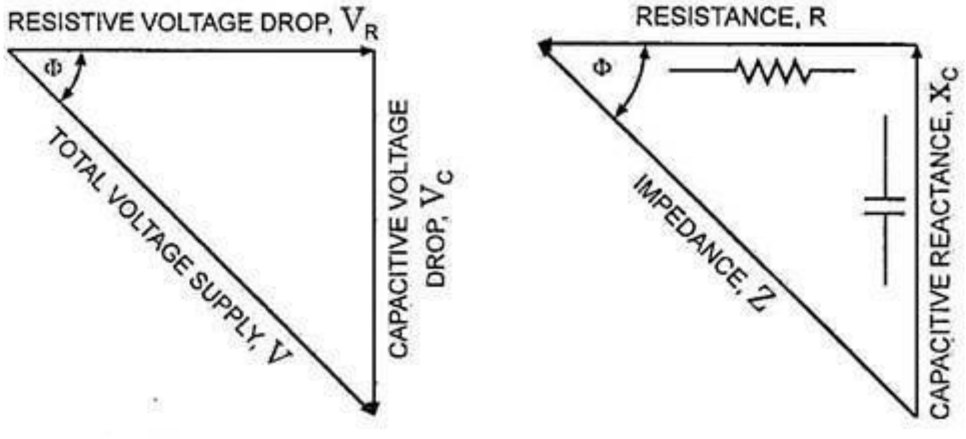
$$v = V_{max} \sin \omega t$$

Then instantaneous current will be expressed as:

$$i = I_{max} \sin(\omega t + \phi)$$

And power consumed by the circuit is given by:

$$P = VI \cos\phi$$



(a) Voltage Triangle *(b) Impedance Triangle*

Fig. 4.19

Voltage triangle and impedance triangle Fig. 4.19 are shown in Figs. 4.19 (a) and 4.19 (b) respectively.

6. Apparent Power, True Power, Reactive Power and Power Factor:

The product of rms values of current and voltage, VI is called the apparent power and is measured in volt-amperes or kilo-volt amperes (kVA).

The true power in an ac circuit is obtained by multiplying the apparent power by the power factor and is expressed in watts or kilo-watts (kW).

The product of apparent power, VI and the sine of the angle between voltage and current, sin ϕ is called the reactive power. This is also known as wattless power and is expressed in reactive volt-amperes or kilo-volt amperes reactive (kVA R).

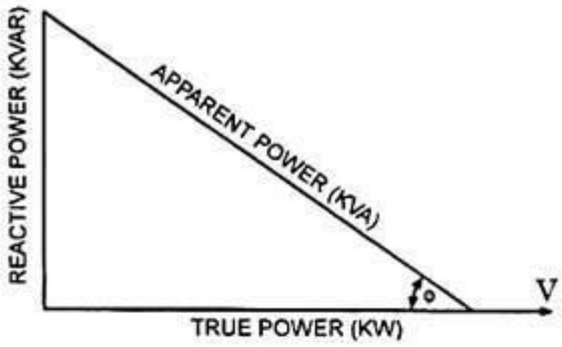
$$\textit{i.e.}\text{ Apparent power, } S = VI \text{ volt-amperes or } \frac{VI}{1{,}000} \text{ kVA}$$

$$\text{True power, } P = VI\cos\Phi \text{ watts or } \frac{VI\cos\Phi}{1{,}000} \text{ kW}$$

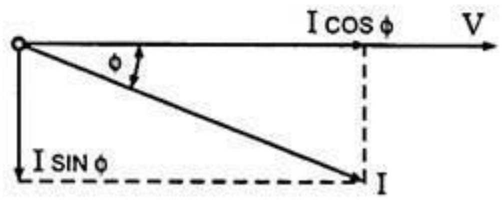
$$\text{Reactive power, } Q = VI\sin\Phi \text{ VAR or } \frac{VI\sin\Phi}{1{,}000} \text{ kVAR}$$

$$\text{and kVA} = \sqrt{(kW)^2 + (kVAR)^2}$$

The above relations can easily be followed by referring to the power diagram shown in Fig. 4.7 (a).



(*a*) *Power Triangle*



(*b*)

Fig. 4.7

Power factor may be defined as:

(i) Cosine of the phase angle between voltage and current,

(ii) The ratio of the resistance to impedance, or

(iii) The ratio of true power to apparent power.

The power factor can never be greater than unity. The power factor is expressed either as fraction or as a percentage. It is usual practice to attach the word ‘lagging’ or ‘leading’ with the numerical value of power factor to signify whether the current lags behind or leads the voltage.

CONCEPT OF REACTANCE, IMPEDANCE, SUSCEPTANCE AND ADMITTANCE:

Reactance is essentially inertia against the motion of electrons. It is present anywhere electric or magnetic fields are developed in proportion to applied voltage or current, respectively; but most notably in capacitors and inductors. When alternating current goes through a pure reactance, a voltage drop is produced that is 90^{o} out of phase with the current. Reactance is mathematically symbolized by the letter “X” and is measured in the unit of ohms (Ω).

Impedance is a comprehensive expression of any and all forms of opposition to electron flow, including both resistance and reactance. It is present in all circuits, and in all components. When alternating current goes through an impedance, a voltage drop is produced that is somewhere between 0^{o} and 90^{o} out of phase with the current. Impedance is mathematically symbolized by the letter “Z” and is measured in the unit of ohms (Ω), in complex form

Admittance is also a complex number as impedance which is having a real part, Conductance (G) and imaginary part, Susceptance (B).

$$Y = G + jB$$
$$Y \rightarrow Admittance\ in\ Siemens$$
$$G \rightarrow Conductance\ in\ Siemens = \frac{R}{R^2 + X^2}$$
$$B \rightarrow\ Susceptance\ in\ Siemens = -\frac{X}{R^2 + X^2}$$

(it is negative for capacitive susceptance and positive for inductive susceptance)

$$j^2 = -1$$
$$|\,Y\,| = \sqrt{G^2 + B^2} = \frac{1}{\sqrt{R^2 + X^2}}$$
$$\angle Y = arctan\left(\frac{B}{G}\right) = arctan\left(-\frac{X}{R}\right)$$

Susceptance (symbolized *B***)** is an expression of the ease with which alternating current (**AC**) passes through a capacitance or inductance